Acute Kidney Injury in Children

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Introduction

Acute kidney injury (AKI) previously known as acute renal failure (ARF), is an important emergency where prompt and appropriate management is life-saving. AKI usually occurs in patients with previously normal renal function but may occasionally be superimposed on preexisting renal disease (acute-on-chronic renal failure). The incidence of AKI in pediatric intensive care unit (PICU) is around 30–40% with mortality rates of 40–50%.

Nomenclature and Classification

- ☑ Acute kidney injury is defined as an increase in serum creatinine by ≥0.3 mg/dL within 48 hours; or an increase in serum creatinine to ≥1.5 times baseline, which is known or presumed to have occurred within last 7 days; or a urine volume <0.5 mL/kg/h for 6 hours.
- ☑ Acute kidney injury is further classified into three stages based on rise of serum creatinine or change in urine volume **(Table 1)**. It is recommended to use Kidney Disease: Improving Global Outcomes (KDIGO) classification for an early diagnosis of AKI so that measures are taken to prevent the progression of the condition.

Nomenclature and Classification

TABLE 1: Classification of acute kidney injury (KDIGO classification).

AKI severity	*Serum creatinine criteria*	*Urine output criteria*
Stage I	1.5–1.9 times baseline Or ≥0.3 mg/dL increase	<0.5 mL/kg/h for 6–12 hours
Stage II	Increase ≥2–2.9 times baseline	<0.5 mL/kg/h for ≥12 hours
Stage III	3.0 times baseline Or Increase in serum creatinine to ≥4.0 mg/dL Or Initiation of renal replacement therapy Or In patients <18 years, decrease in eGFR to <35 mL/min per 1.73 m^2	<0.3 mL/kg/h for 24 hours or anuria for 12 hours

(AKI: acute kidney injury; eGFR: estimated glomerular filtration rate; KDIGO: Kidney Disease: Improving Global Outcomes)

Causes

The etiology of AKI has conventionally been classified as prerenal, intrinsic renal, or postrenal **(Box 1)**. Prerenal AKI occurs due to inadequate systemic and/or renal circulation, due to either systemic hypovolemia or renal hypoperfusion. Both pre- and postrenal categories can, if prolonged, lead to intrinsic renal failure.

BOX 1: Causes of AKI.

Prerenal

- ☑ Hypovolemia (dehydration, blood loss, and diabetic ketoacidosis)
- ☑ Third space losses (septicemia and nephrotic syndrome)
- ☑ Congestive heart failure
- ☑ Perinatal asphyxia
- ☑ Drugs (ACE inhibitors, NSAIDs, and diuretics)

Intrinsic

Acute tubular necrosis

- ☑ Prolonged prerenal insult (see above)
- ☑ *Medications:* Aminoglycoside, radiocontrast, and NSAIDs
- ☑ *Exogenous toxins:* Diethylene glycol and methanol
- ☑ Intravascular hemolysis and hemoglobinuria
- ☑ Snake bite
- ☑ Tumor lysis syndrome

Contd...

Causes

Contd...

Hemolytic uremic syndrome: Diarrhea associated (D+) and atypical (D-) forms
Glomerulonephritis (GN)
- ☑ Postinfectious GN
- ☑ *Systemic disorders:* SLE, Henoch–Schönlein syndrome and microscopic polyangiitis
- ☑ Membranoproliferative GN

Interstitial nephritis (drug-induced and idiopathic)
Bilateral renal vessel occlusion (arterial and venous)

Postrenal
- ☑ Posterior urethral valves and urethral stricture
- ☑ Bilateral pelviureteric junction obstruction
- ☑ Ureteral obstruction (stenosis, stone, and ureterocele)
- ☑ Neurogenic bladder

(ACI: angiotensin-converting enzyme; AKI: acute kidney injury; NSAIDs: nonsteroidal anti-inflammatory drugs; SLE: systemic lupus erythematosus)

Clinical Features

Presenting symptoms of AKI include oliguria with peripheral or pulmonary edema, suggesting fluid overload. Patients may have altered sensorium and convulsions due to advanced uremia, dyselectrolytemia, or hypertensive encephalopathy. The breathing may be rapid and deep due to acidosis. Features that suggest an underlying cause are presented in **Table 2**. Children with AKI due to acute interstitial nephritis, aminoglycoside toxicity, and perinatal asphyxia are often nonoliguric.

TABLE 2: Clinical features.

Clinical features	***Likely diagnosis***
Edema, hematuria, and hypertension	Acute glomerulonephritis
Dysentery, pallor, and jaundice	HUS
History of fluid loss with severe dehydration	ATN
Sudden passage of dark urine, pallor, and jaundice	Intravascular hemolysis
Interrupted urinary stream and palpable bladder	Obstructive uropathy
Abdominal colic, hematuria, and dysuria	Urinary tract calculi
Altered sensorium and seizures	Uremic encephalopathy
Acidotic breathing and pulmonary edema	Complications of AKI

(AKI: acute kidney injury; ATN: acute tubular necrosis; HUS: hemolytic uremic syndrome)

Investigations

Investigations in patients with AKI are given in **Box 2**.

BOX 2: Investigations in patients with AKI.

Blood

- ☑ Complete blood counts
- ☑ KFT
- ☑ Electrolytes (Na, K, and Ca)
- ☑ Venous blood gas (pH and bicarbonate)

Urine

- ☑ Urinalysis; culture (if symptoms of urinary infection)
- ☑ Sodium, osmolality, fractional excretion of sodium (if available to differentiate prerenal from intrinsic AKI)

Radiology

- ☑ Chest X-ray (for fluid overload and cardiomegaly)
- ☑ Ultrasonography (identify obstruction and dilatation)
- ☑ ECG for hyperkalemia

Investigations to determine cause

- ☑ Peripheral smear examination, platelet, and reticulocyte count; blood and LDH levels; and stool culture (suspected hemolytic uremic syndrome)
- ☑ Peripheral smear/RMAT—malaria
- ☑ Leptospiral serology/microscopic agglutination test (gold standard)—leptospirosis
- ☑ Blood culture—sepsis
- ☑ Blood ASO, complement (C3), antinuclear antibody (ANA), antineutrophil cytoplasmic antibody (ANCA) (suspected acute and rapidly progressive GN)
- ☑ Doppler ultrasonography (suspected arterial or venous thrombosis)
- ☑ Renal biopsy in RPGN or nonresolving AKI
- ☑ Micturating cystourethrogram (suspected obstruction)

(AKI: acute kidney injury; ASO: antistreptolysin O; ECG: electrocardiogram; GN: glomerulonephritis; KFT: kidney function test; LDH: lactate dehydrogenase; RMAT: rapid malarial antigen test; RPGN: rapidly progressive glomerulonephritis)

Management of AKI includes the management of complications and treatment of the specific underlying cause **(Table 3)**.

TABLE 3: Management of complications.

Complications	***Management***	***Others***
Fluid overload	☑ AKI regimen—insensible losses to be replaced by 5% Dextrose ☑ UO to be replaced by NS ☑ Modify according to Na level	☑ Replace other losses ☑ Consider dialysis ☑ 0.5–1% weight loss per day
Pulmonary edema	☑ O_2/IV furosemide (2–4 mg/kg) ☑ Respiratory support—HFNC/NIV/MV according to the patient condition	☑ Chest X-ray ☑ Lung ultrasound—B line ☑ IVC assessment and ejection fraction ☑ Monitor by CVP line ☑ Dialysis (for fluid removal)
Hypertension	☑ Symptomatic – Labetalol infusion @ 0.25–1 mg/kg/h – IV furosemide 2–4 mg/kg ☑ Asymptomatic—oral nifedipine/amlodipine 0.3–0.5 mg/kg	☑ NTP—0.5–8 µg/kg/min (not to be given for >48 hours due to risk of toxicity) ☑ Maintenance with amlodipine/hydralazine
Metabolic acidosis	IV or oral $NaHCO_3$	To monitor for fluid overload and hypernatremia
Hyperkalemia	☑ Stop all potassium in IVF and medications ☑ IV 10% Ca gluconate 1 mL/kg over 5–10 minutes if ECG changes ☑ Salbutamol (2.5–5 mg) nebulization. Can be repeated after 20 minutes ☑ Neutralizing glucose insulin drip—dextrose 0.5–1 g/kg and insulin 0.1–0.2 U/kg over 30 minutes. Can be repeated after 30 minutes ☑ $NaHCO_3$—1–2 mL/kg over 15–20 minutes if associated with metabolic acidosis ☑ Potassium binding resins—sodium polystyrene 1 g/kg (oral/NG/rectal) maximum 30 g	☑ Monitor blood glucose for hypoglycemia ☑ Continuous ECG monitoring ☑ Repeat potassium level
Anemia	PCV transfusion at 5–10 mL/kg	Monitor for fluid overload
Hyper-phosphatemia	Phosphate binders such as calcium carbonate, and sevelamer hydrochloride (only if significant)	

(AKI: acute kidney injury; CVP: central venous pressure; ECG: electrocardiogram; HFNC: high-flow nasal cannula; IV: intravenous; IVC: inferior vena cava; MV: mechanical ventilation; NG: nasogastric; NIV: noninvasive ventilation; NS: normal saline; PCV: packed cell volume; UO: urine output)

Nutrition

- ☑ Patients with AKI are usually catabolic and have increased metabolic needs. Adequate nutritional support is desirable with maximization of caloric intake. However, volume restriction necessary during the oliguric phase often imposes limitations. A diet containing 0.8–1.2 g/kg of protein in infants and 0.6–0.8 g/kg in older children and a minimum of 50–60 Cal/kg should be given. The latter requirement can be met by adding liberal amounts of carbohydrates and fats to the diet. Once dialysis is initiated, dietary fluid and electrolyte restrictions can be made more liberal.

Kidney Replacement Therapy

Indications of kidney replacement therapy (KRT) are given in **Table 4**.

TABLE 4: Indications of kidney replacement therapy (KRT).

Indications	***Features***
Fluid overload	☑ Most common indication ☑ Determines the outcome ☑ >15% fluid overload resistant to diuretics
Metabolic acidosis	pH<7.2 despite bicarbonate therapy
Refractory hyperkalemia	K^+ >6.0 or electrocardiogram (ECG) changes despite medical management
Hyponatremia/hypernatremia	Symptomatic
Uremia	Urea >160–200 mg/dL, encephalopathy
Create space for more fluid	Blood products, drugs, and nutrition
Removal of dialyzable toxins	Salicylate poisoning and phenobarbitone

- ☑ Acute kidney injury requiring dialysis can be managed with a variety of modalities, including peritoneal dialysis (PD), intermittent hemodialysis (HD), and continuous renal replacement therapy (CRRT) (continuous venovenous hemofiltration or hemodiafiltration). The choice of dialysis modality to be used in managing a specific patient is influenced by several factors, including the goals of dialysis, the unique advantages and disadvantages of each modality, and institutional resources **(Table 5)**. The initial KRT of choice in sick and unstable patients is often PD. It is popular because of the ease of initiation and effectiveness in children of all ages, including neonates.

Kidney Replacement Therapy

TABLE 5: KRT modalities.

Modality	*Potential setting in AKI*	*Advantages*	*Disadvantages*
IHD	Hemodynamically stable	☑ Rapid removal of toxins ☑ Reduced exposure to anticoagulants ☑ Lower cost than CRRT	☑ Hypotension with rapid fluid removal ☑ Dialysis disequilibrium
CRRT	Hemodynamically unstable	☑ Continuous removal of toxins ☑ Hemodynamic stability ☑ Easy control of fluid balance ☑ No risk of increased ICP	☑ Slower clearance of toxins ☑ Need for prolonged anticoagulation ☑ Patient immobilization ☑ Increased cost
PD	☑ Hemodynamically unstable ☑ Coagulopathy ☑ Difficult access ☑ Under resourced region	☑ Technically simple ☑ No anticoagulation ☑ Hemodynamic stability ☑ Lower cost ☑ No need for vascular access	☑ Poor clearance ☑ Protein loss ☑ Risk of peritonitis ☑ No control on rate of fluid removal ☑ Hyperglycemia

(AKI: acute kidney injury; CRRT: continuous renal replacement therapy; ICP: intracranial pressure; IHD: intermittent hemodialysis; KRT: kidney replacement therapy; PD: peritoneal dialysis)

Further Reading

- ☑ Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO Clinical Practice Guideline for Acute Kidney Injury. Kidney inter. 2012;2:1-138.
- ☑ Levey AS, Eckardt KU, Dorman NM, Christiansen SL, Hoorn EJ, Ingelfinger JR, et al. Nomenclature for kidney function and disease: report of a Kidney Disease: Improving Global Outcomes (KDIGO) Consensus Conference. Kidney Int. 2020;97(6):1117-29.
- ☑ Sutherland SM, Kwiatkowski DM. Acute kidney injury in children. Adv Chronic Kidney Dis. 2017;24(6):380-7.

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Immune Thrombo-cytopenic Purpura (ITP)

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Immune Thrombo-cytopenic Purpura (ITP)

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Nomenclature

Earlier ITP was referred to as immune thrombocytopenic purpura; however, according to recent guidelines it is accepted as "immune thrombocytopenia".

Definitions

TABLE 1: Treatment of different categories of ITP.

Newly diagnosed ITP/ acute immune thrombo-cytopenic purpura (ITP)	Lasting <3 months
Persistent ITP	Lasting 3–12 months
Chronic ITP	Lasting beyond 12 months
Primary ITP	A condition in which platelet count is <100,000/mm^3 in absence of other disorders or conditions associated with thrombocytopenia
Secondary ITP	When it is caused due to: ☑ Infections—human immunodeficiency virus (HIV), hepatitis C virus (HCV), *Helicobacter pylori*, *Cytomegalovirus* (CMV), and varicella zoster ☑ Autoimmune conditions—systemic lupus erythematosus (SLE), antiphospholipid antibody (APLA) syndrome, and Evans syndrome ☑ Primary immunodeficiency disorder—common variable immunodeficiency (CVID) ☑ Drugs—valproate and heparin ☑ Miscellaneous—lymphoproliferative disorder